

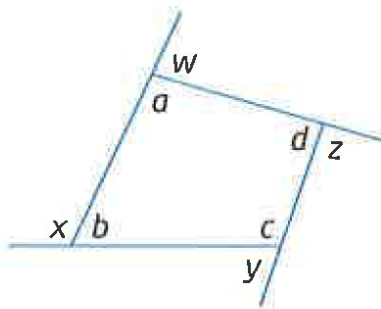
Section 7.2 – Angle Relationships in Quadrilaterals

MPM1D

Angle Relationships in Quadrilaterals

The sum of the **interior** angles of a quadrilateral is 360 degrees.

The sum of the **exterior** angles of a quadrilateral is also 360 degrees.



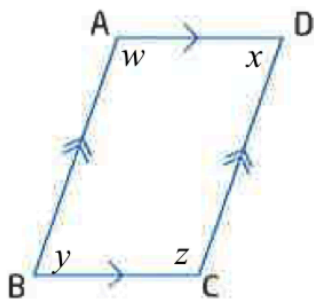
Interior angles:
 $a + b + c + d = 360^\circ$

Exterior angles:
 $w + x + y + z = 360^\circ$

Angle Relationships in Parallelograms

Adjacent angles in a parallelogram are supplementary (add to 180).

Opposite angles in a parallelogram are equal.



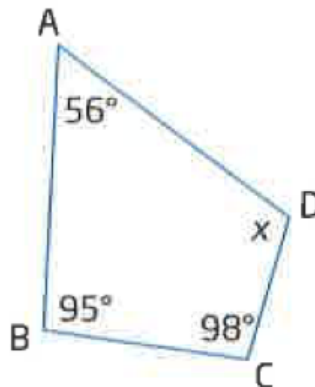
Adjacent angles:

$$\begin{aligned}w + x &= 180 \\w + y &= 180 \\y + z &= 180 \\z + x &= 180\end{aligned}$$

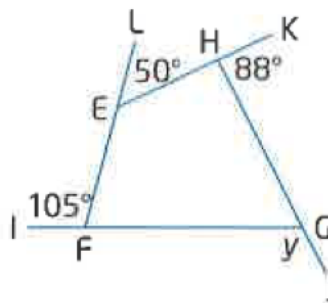
Opposite angles:

$$\begin{aligned}w &= z \\x &= y\end{aligned}$$

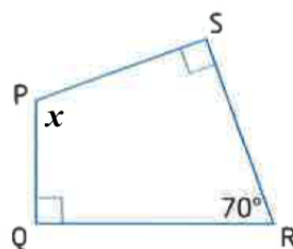
Example 1: Find the measure of the unknown angle



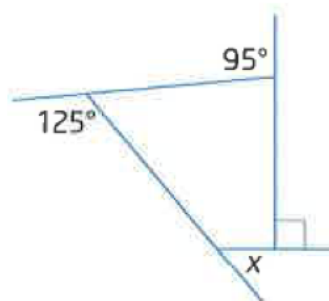
Example 2: Find the measure of the unknown angle



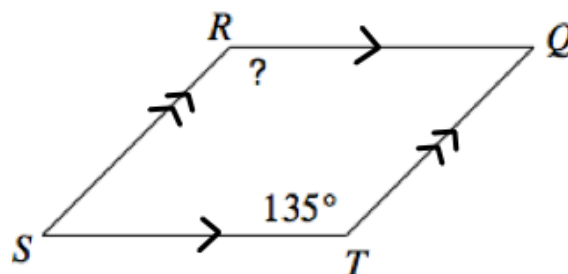
Example 3: Find the measure of the unknown angle



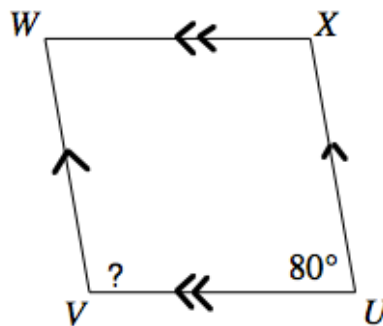
Example 4: Find the measure of the unknown angle



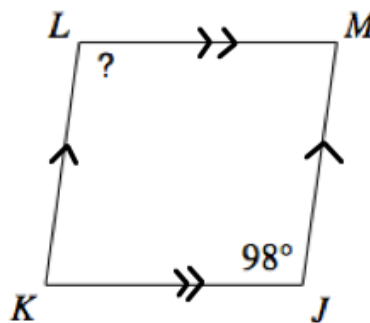
Example 5: Find the measure of the unknown angle



Example 6: Find the measure of the unknown angle



Example 7: Find the measure of the unknown angle



Example 8: Find the measure of the unknown angle

